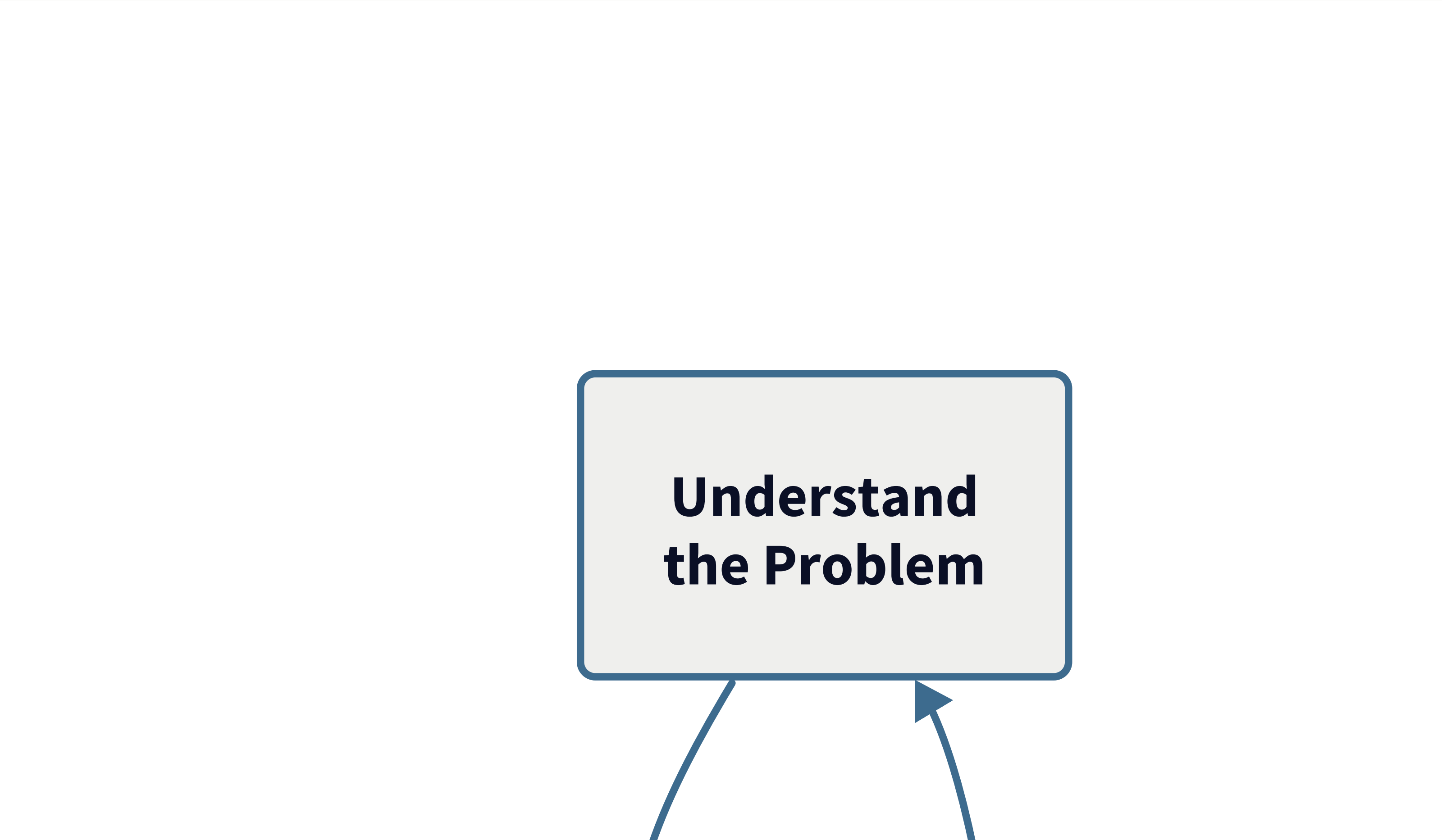
Day 04

The Modeling Project

MSU REU Machine Learning Short Course

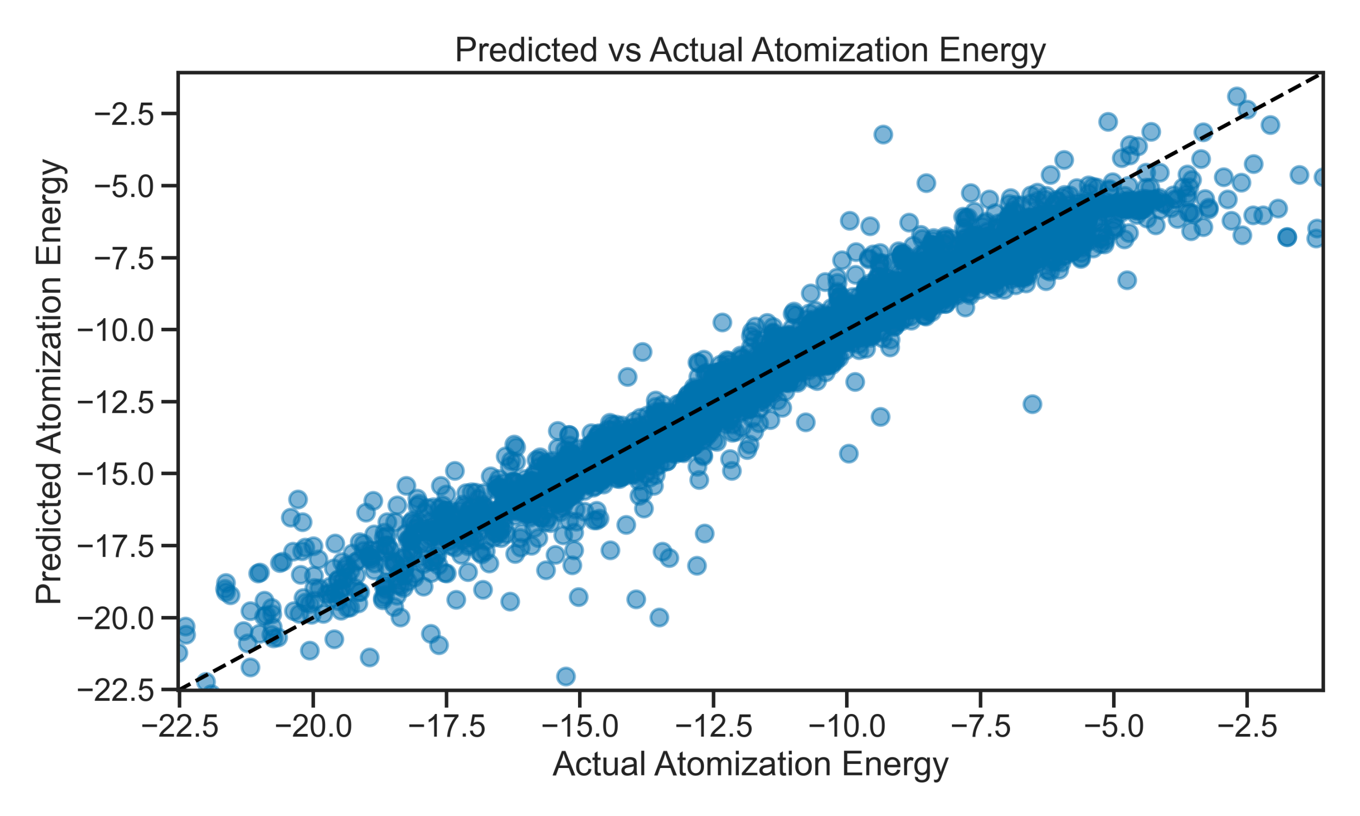
The Full ML Loop

You've seen each piece. Today you run the whole thing yourself.



Today's Dataset — Molecular Atomization Energy

Predict the energy required to pull a molecule apart, from its atomic structure.



1275 features from the Coulomb matrix representation of each molecule. Most of them are noise.

Start with a terrible model

Build the simplest possible model first.

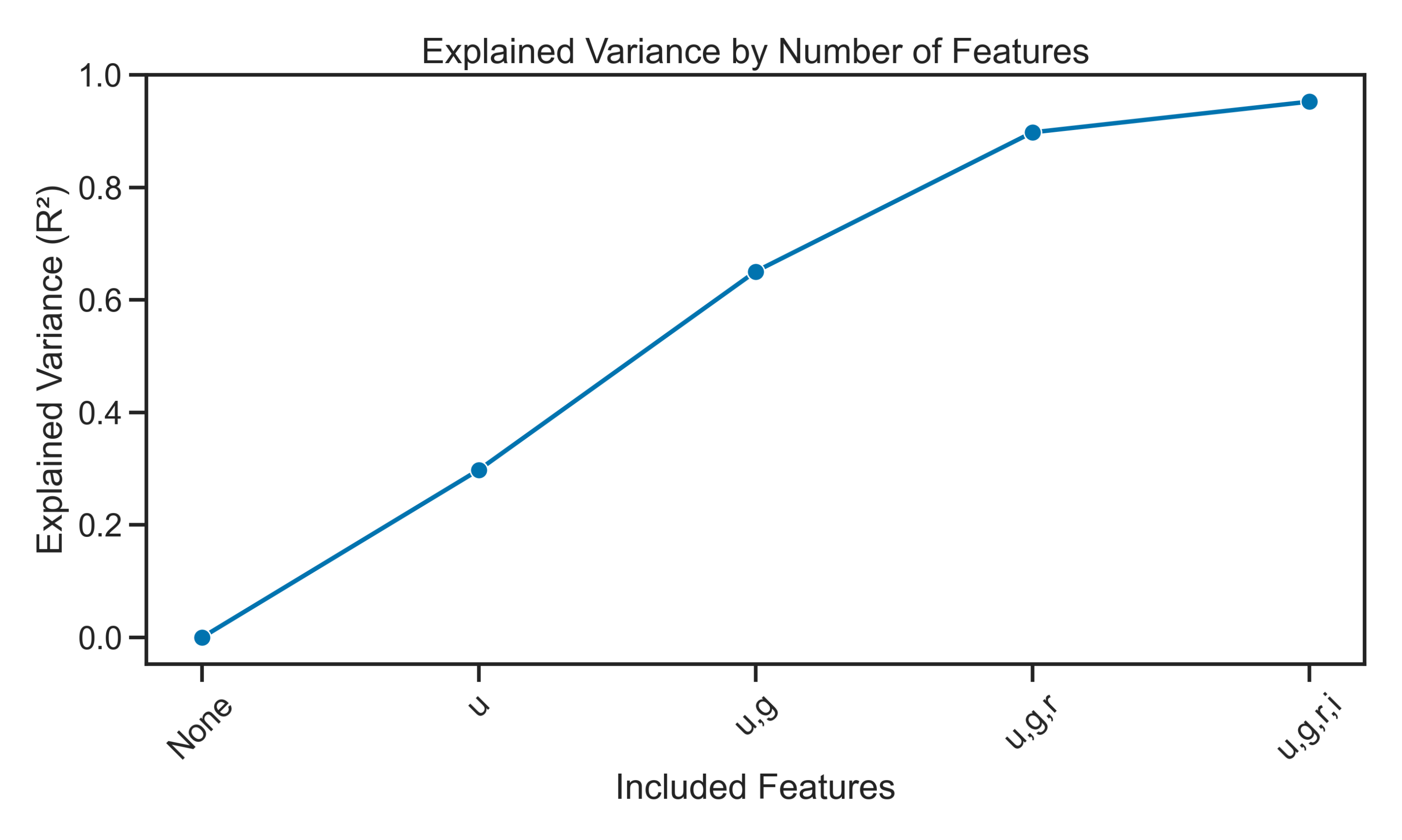
```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LinearRegression()
model.fit(X_train, y_train)
print(r2_score(y_test, model.predict(X_test)))   # probably bad
```

Then explore the data to understand why it's bad.

Hint: plot a histogram of each feature. What do you notice?

Feature selection — less is more

Not all 1275 features carry signal. Recursive Feature Elimination finds which ones to keep.



See: [Recursive Feature Elimination]()

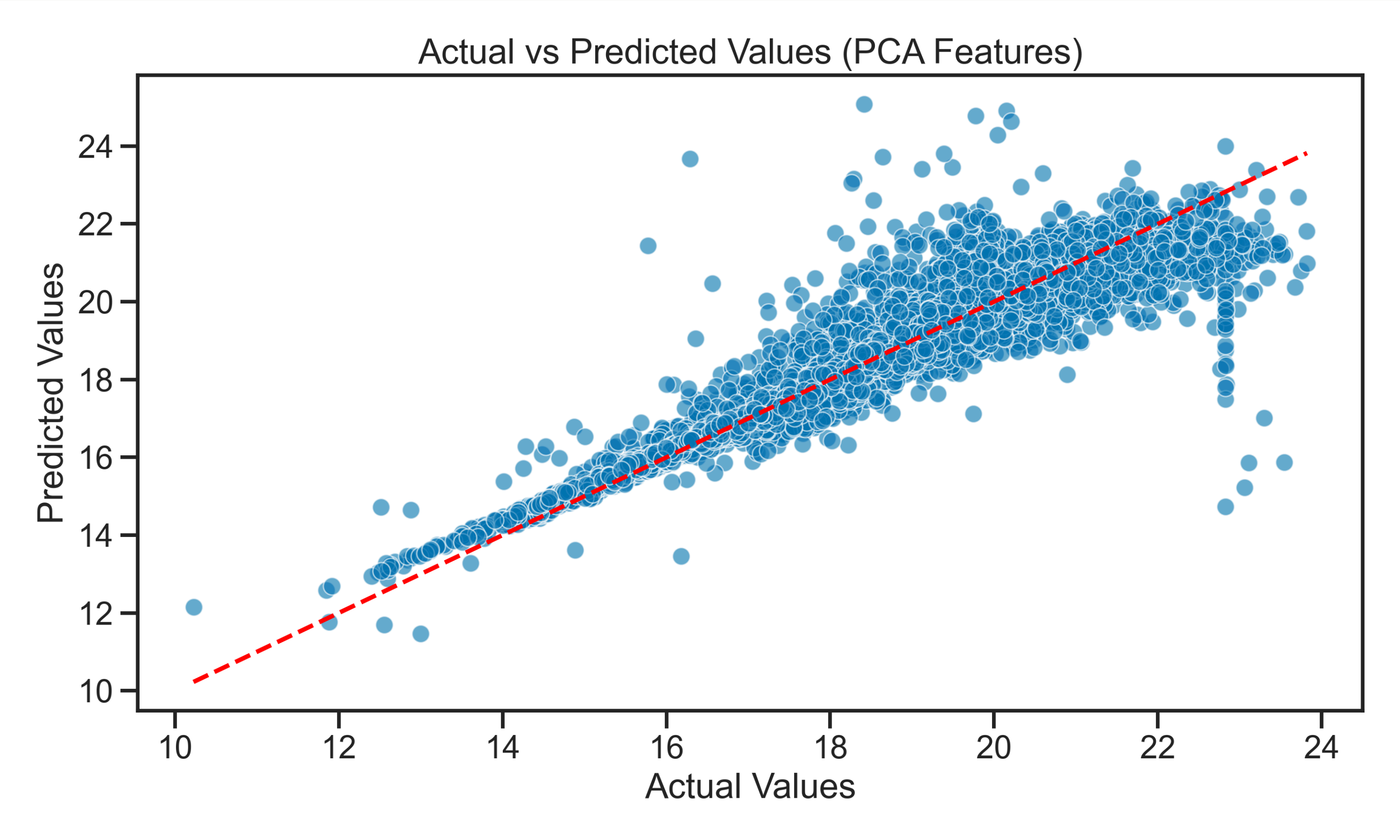
Cross-validation — how confident are you?

One train/test split gives you one number. Run the model many times to get a distribution.

Each fold takes a turn as the test set.
This gives 5 independent estimates of model performance.

Dimensionality reduction — PCA

1275 correlated features → a small number of uncorrelated components that capture most of the variance.



See: [Principal Component Analysis]

Today's Activity

Work through **Activity 04: Modeling Project** — open-ended by design.

1. Load the molecular energy dataset
2. Build an initial model — expect it to be bad
3. Explore: why is it bad? Remove useless features.
4. Re-evaluate. Does it improve?
5. Implement cross-validation to estimate performance with uncertainty.

> **Notes:** Cross-Validation · RFE · PCA
>
> Use everything from Days 1–3. Ask questions. Improve iteratively.